

Commensurate to incommensurate magnetic phase transition in Honeycomb-lattice pyrovanadate $\text{Mn}_2\text{V}_2\text{O}_7$

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We have synthesized single crystalline sample of $\text{Mn}_2\text{V}_2\text{O}_7$ using floating zone technique and investigated the ground state using magnetic susceptibility, heat capacity and neutron diffraction. Our magnetic susceptibility and heat capacity reveal two successive magnetic transitions at $T_{N1} = 19$ K and $T_{N2} = 11.8$ K indicating two distinct magnetically ordered phases. The single crystal neutron diffraction study shows that in the temperature (T) range $11.8 \text{ K} \leq T \leq 19 \text{ K}$ the magnetic structure is commensurate with propagation vector $k_1 = (0, 0.5, 0)$, while upon lowering temperature below $T_{N2} = 11.8$ K an incommensurate magnetic order emerges with $k_2 = (0.38, 0.48, 0.5)$ and the magnetic structure can be represented by cycloidal modulation of the Mn spin in ac -plane. We are reporting this commensurate to incommensurate transition for the first time. We discuss the role of the magnetic exchange interactions and spin-orbital coupling on the stability of the observed magnetic phase transitions.

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I. INTRODUCTION

Bulk crystals are inherently three dimensional; however, they may consist of magnetic ions whose spins interact only along one or two certain crystallographic directions. Such type of compounds are known as low dimensional magnets which have aroused considerable attention in solid state chemistry and physics due to their non-typical behaviour which mainly deal with electrical and magnetic properties. Two dimensional honeycomb-lattice systems are one of the interesting low dimensional magnetic systems which have been intensively investigated because of novel ground states induced by frustration and quantum fluctuations.^{1–5} Magnetic order can be obtained on a regular two dimensional lattice assuming antiferromagnetic interactions between adjacent spins;⁶ or on a bilayer lattice with frustrating interlayer interactions.⁷ In recent times, these honeycomb lattice systems have been extensively investigated both theoretically and experimentally as potential candidate for Kitaev quantum spin liquid state, for example Na_2IrO_3 , $\alpha\text{-RuCl}_3$.^{8,9} Moreover, superconductivity has also been observed in pnictide SrPtAs with honeycomb lattice.¹⁰

The title compound $\text{Mn}_2\text{V}_2\text{O}_7$ is a member of the family of transition metal-vanadium oxides $\text{M}_2\text{V}_2\text{O}_7$ ($\text{M} = \text{Cu}, \text{Ni}, \text{Co}, \text{Mn}$) which have attracted much interest due to their rich structural features and magnetic properties.^{11–19} $\text{Mn}_2\text{V}_2\text{O}_7$ is composed of magnetic Mn^{2+} ($3d^5$, $S = 5/2$) and non-magnetic V^{5+} ($3d^0$, $S = 0$) ions and was reported to possess quasi two dimensional distorted honeycomb lattice.^{20,21} It exhibits two different structural phases; namely β - phase (above ≈ 310 K) with monoclinic ($C2/m$) symmetry and α - phase (below ≈ 284 K) with triclinic ($P\bar{1}$) symmetry. As temperature decreases, the high temperature monoclinic symmetry

of $\text{Mn}_2\text{V}_2\text{O}_7$ reduces to triclinic one and this structural transition is completely reversible with reasonable thermal hysteresis, in line with the martensitic-like first order nature of this structural transition involving lattice distortion but without atomic exchange.^{18,21} Depending on the synthesis procedure these transition temperatures vary because of the change in O_2 pressure and cooling rate. The honeycomb networks of Mn-atoms in the β phase are parallel to (001) plane and those in the α phase are parallel to the (110) plane. Further the sample undergoes paramagnetic to antiferromagnetic transition below around 19 K. It is noted that such properties of $\text{Mn}_2\text{V}_2\text{O}_7$ were mainly based on macroscopic characterization methods on polycrystalline as well as flux grown single crystalline samples. But a detailed microscopic model of the magnetism has not yet been developed.

So far there are few reports on the single crystal growth of $\text{Mn}_2\text{V}_2\text{O}_7$ where only flux growth technique has been followed.^{22,23} The crystals obtained are small and moreover the grown crystals may incorporate traces of the molten flux or the crucible materials, which is highly undesirable. Keeping this view in mind, we decided to grow $\text{Mn}_2\text{V}_2\text{O}_7$ crystals using the traveling solvent floating zone (TSFZ) method associated with an optical image furnace. $\text{Mn}_2\text{V}_2\text{O}_7$ melts congruently upon heating at $(1080 \pm 3)^\circ \text{C}$.²⁴ In order to understand the structural and magnetic properties of the ground state of $\text{Mn}_2\text{V}_2\text{O}_7$ at the macroscopic as well as microscopic level, we carried out x-ray diffraction (XRD), magnetization and heat capacity measurement along with the single crystal neutron diffraction study.

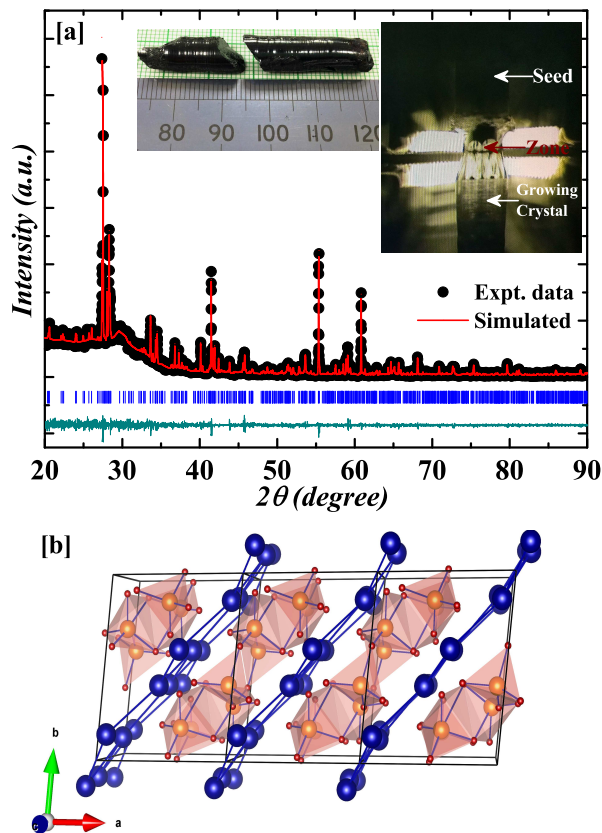


FIG. 1. (a) Main panel shows the XRD pattern of crushed single crystal measured at $T = 100$ K. Black scattered points and red solid line represent respectively the experimental data and the simulated curve obtained from Rietveld refinement. Blue ticks indicate the Bragg positions for triclinic $P\bar{1}$ structure and the cyan line shows the difference between experimental and calculated pattern. Right inset shows the stable solvent zone formation during growing, while left inset depicts the picture of the as grown crystal. (b) depicts the perspective view of the crystal structure of $\text{Mn}_2\text{V}_2\text{O}_7$ at 100 K where blue, yellow and red spheres represent Mn, V and O atoms respectively.

II. EXPERIMENTAL DETAILS

The single crystalline sample of $\text{Mn}_2\text{V}_2\text{O}_7$ has been grown in a four-mirror optical floating zone furnace from Crystal System Corporation, Japan. The furnace consists of four halogen lamps. Every lamp takes the focus position of an ellipsoidal reflector whereas the second focus of the reflectors coincide at a point inside growth chamber on the vertical axis of the furnace. A molten zone between the polycrystalline feed rod and the seed rod is formed at this common focal point where the energy flux of the lamps converge after reflecting from the reflectors. The feed rod is suspended from the upper shaft with a nickel wire while the seed rod is clamped to the lower shaft using an alumina holder. Polycrystalline sample of $\text{Mn}_2\text{V}_2\text{O}_7$ was prepared by standard solid state reaction route in air. Highly pure MnO_2 and

V_2O_5 were mixed thoroughly in a stoichiometric ratio in an agate mortar. The mixture was pressed into pellets and sintered at 600°C for one week with several intermediate grindings. The phase analysis of the final product was performed using powder x-ray diffraction (Miniflex). The single phase powder of $\text{Mn}_2\text{V}_2\text{O}_7$ was filled in rubber tube and cold pressed under 700 bar isostatic pressure to obtain uniform rod. The rods formed in this process are 5 mm in diameter and 70 - 90 mm in length. Then the rods were sintered in the platinum boat at the temperature 900°C overnight to obtain very dense and homogeneous rod. This process ensures the homogeneous melting of the feed rod during growth process. A part about 20 mm long was cut from the rod and used as the seed rod.

Three different growth experiments were performed with different oxygen pressure and growth speed employed; (i) 60% O_2 + 40% Ar and growth speed 5 mm/hr (ii) 100 % O_2 flow ($P = 7$ bar) and growth speed 5 mm/hr and (iii) 100 % O_2 flow ($P = 7$ bar) and growth speed 1 mm/hr. The growth process is initiated by fusing the bottom end of the feed rod inside the optical furnace. In our case 300 watt lamps have been installed. The upper and lower shafts of the furnace were rotated in opposite directions with rotation speed 30 rpm to improve the temperature and compositional homogeneity of the float zone. In the first two attempts, upper shaft was translated at a speed of 1 mm/hr and in the third attempt it is translated at a speed of 0.5 mm/hr in order to control the steady state of floating zone. In the first attempt molten zone was not fully stable and formation of bubbles were observed near the upper edge of the zone possibly due to the reduction of Mn ions. Then we decided to use 100 % O_2 pressure to prevent the reduction of Mn^{2+} and with this environment we got significantly stable zone throughout the growth process as shown in the right inset of figure 1(a).

The grown crystals were characterized using powder x-ray diffraction of crushed crystal (Bruker D8 Advance), Laue x-ray diffraction and single crystal neutron diffraction in SXD diffractometer, ISIS neutron and muon facility. The magnetization of the crystals was measured by applying an external field along the three different crystallographic directions using a vibrating sample magnetometer (SQUID VSM, Quantum Design) in the temperature range $T = 2$ to 320 K and applied magnetic field (H) ranging from -70 to +70 kOe. Heat capacity has been measured in Quantum Design PPMS. The x-ray diffraction (XRD) pattern of crushed crystal measured at 100 K is shown in the main panel of figure 1(a). Rietveld refinement on powder XRD data was performed using Fullprof software package and the data is well fitted using triclinic structure with $P\bar{1}$ space group. The crystal structure (shown in figure 1(b)) is consistent with the previously reported literature.

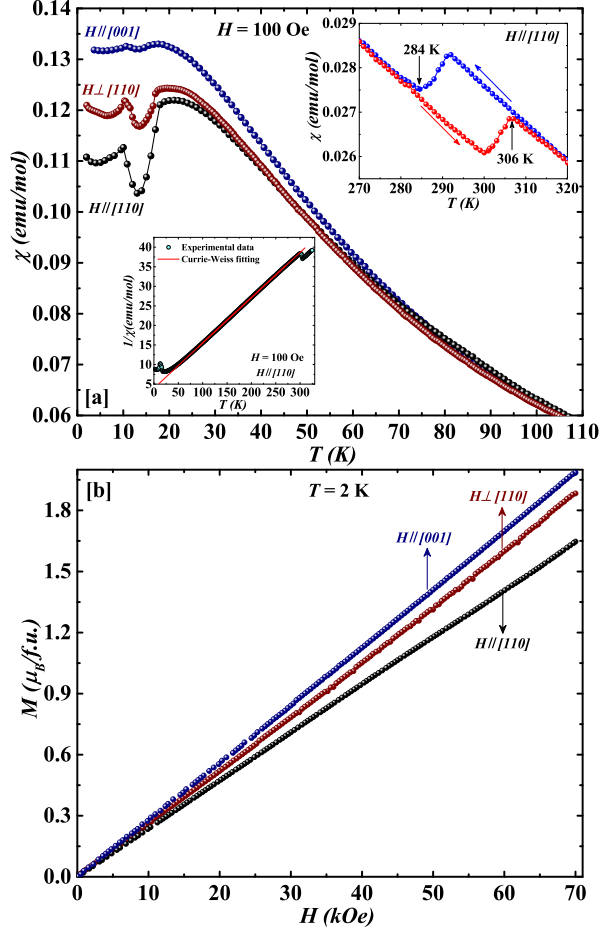


FIG. 2. (a) Main panel shows the temperature dependent magnetization data in the zero field-cooled heating protocol where magnetic field applied along three crystallographic directions: $H \parallel (110)$, $H \perp (110)$ and $H \parallel (001)$. Upper inset depicts the thermal hysteresis observed between 284 K and 306 K on field cooling and field-cooled heating measurements whereas lower inset represents the Curie-Weiss fitting in the paramagnetic region. (b) M versus H curves are displayed measured at $T = 2$ K along three different crystallographic directions.

III. MAGNETIZATION AND HEAT CAPACITY

The main panel of figure 2(a) shows the temperature dependence of magnetic susceptibility (χ) from 2 K to 110 K, which was measured in zero field cooled heating protocol under an applied magnetic field of $H = 100$ Oe along the parallel and perpendicular to (110) plane and parallel to (001) plane of α - $\text{Mn}_2\text{V}_2\text{O}_7$. A clear signature of antiferromagnetic (AFM) transition is observed at $T_{N1} \approx 19$ K which matches quite well with the previously reported data.^{20,21,25} Additionally we found another anomaly at around $T_{N2} = 11.8$ K in all three crystallographic directions. Although the T_{N1} and T_{N2} are present at the same temperatures, but different histories are clearly observed below 19 K dependent on direction

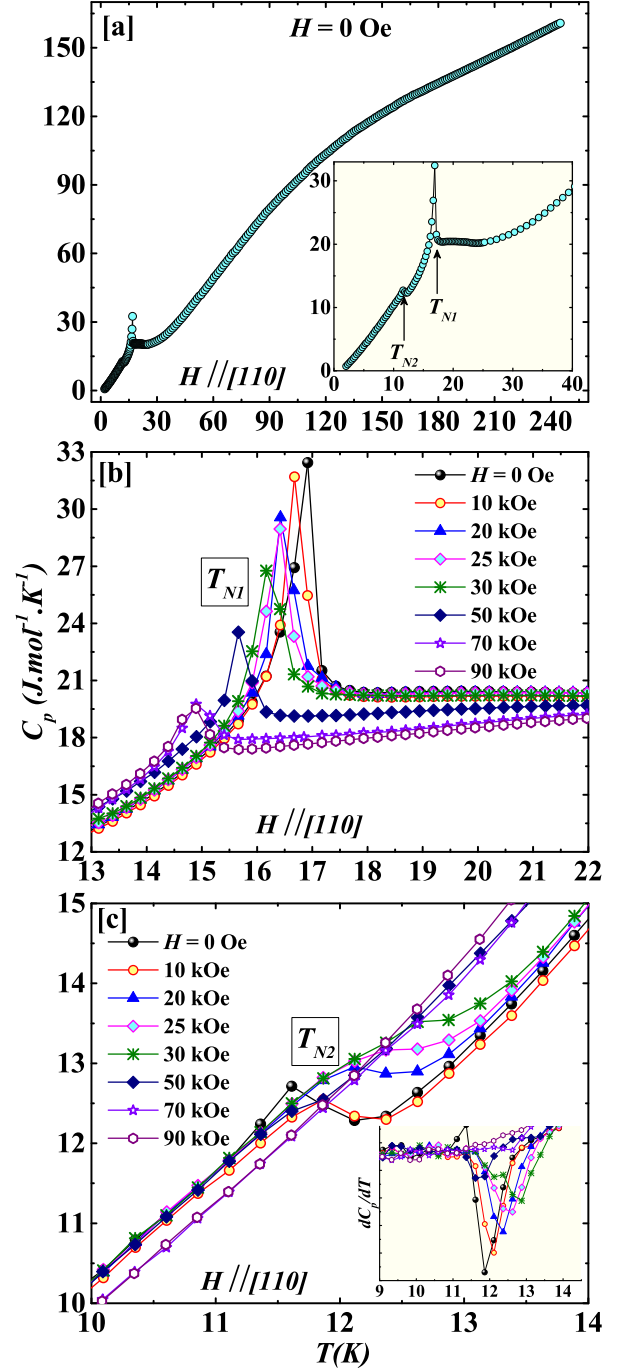


FIG. 3. (a) Main panel shows the temperature dependent heat capacity measured at zero applied magnetic field while inset shows the enlarged view of the low- T regime indicating the two transitions at T_{N1} and T_{N2} . The magnetic field dependence of T_{N1} and T_{N2} are depicted in (b) and (c) respectively. dC_p/dT vs T is plotted in the inset of (c) to show clearly the change in T_{N2} with H .